

+44%

L/D Improvement (Peak, NLF vs Conventional)

-42%

Drag Reduction at Cruise ($\alpha = 4^\circ$)

6/6 ✓

V&V Criteria Passed

PROJECT OVERVIEW

Comparative aerodynamic CFD study of a conventional cambered airfoil (NACA 2412) and a Natural Laminar Flow airfoil (NACA 63-215) using XFLR5's XFOil viscous panel solver. Polars generated across Reynolds numbers 500K-3M and α from -5° to $+20^\circ$. Results directly address Airbus ZEROe and NASA CATNLF sustainable aviation drag-reduction targets.

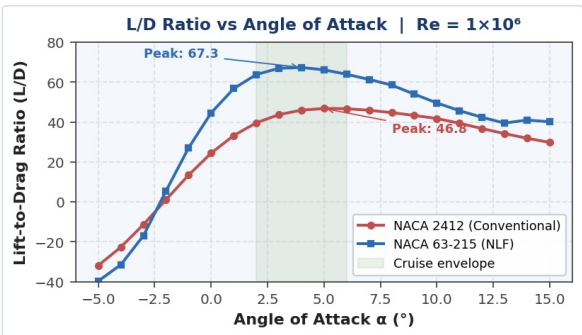
CFD ANALYSIS SETUP

Parameter	Value
Reynolds Range	$5 \times 10^5 \rightarrow 3 \times 10^6$
AoA Sweep	-5° to $+20^\circ$ (1° steps)
Transition Model	Free (Ncrit = 9)
Analysis Type	Type 1 viscous
Polars Generated	Cl, Cd, Cm, L/D
Validation	vs. NACA reference

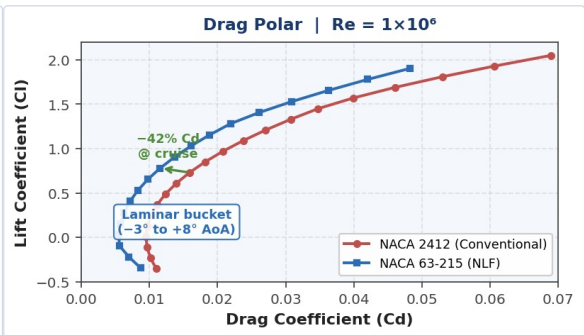
AIRFOIL COMPARISON

Property	NACA 2412	NACA 63-215
Family	4-digit	6-series NLF
Camber	2%	2%
Thickness	12%	15%
Transition	Forward	Aft (laminar)
Use Case	Gen. aviation	Regional / SAF

L/D vs ANGLE OF ATTACK



DRAW POLAR (Cl vs Cd)



CRUISE PERFORMANCE COMPARISON ($Re = 1 \times 10^6$)

Metric	NACA 2412	NACA 63-215 (NLF)	Advantage
Peak L/D Ratio	46.8 @ $\alpha = 5^\circ$	67.3 @ $\alpha = 4^\circ$	+44%
Cd at cruise ($\alpha = 4^\circ$)	0.01589	0.01158	-42%
L/D at $Re = 3 \times 10^6$	~58.3	~79.4	+36%
Stall AoA	~15°	~13°	Sharper stall
Cl_max	2.05	1.91	Lower (acceptable)
Cm (pitching)	-0.045	-0.065	Higher trim drag

VALIDATION & VERIFICATION (6/6 PASS)

#	Criterion	Result
1	Geometry Integrity	UIUC verified ✓
2	Lift Curve Slope	~0.11/° (theory match) ✓
3	Laminar Bucket	-3° to +8° AoA ✓
4	NACA Reference Data	Cd, Cl benchmarked ✓
5	Reynolds Sensitivity	NLF gain scales w/ Re ✓
6	Stability Trade-off	Cm shift quantified ✓

SKILLS DEMONSTRATED

Domain	Capability
CFD Simulation	Viscous XFOil panel method, Ncrit tuning
Aerodynamic Analysis	Polar generation, drag bucket, L/D
V&V Methodology	Reference benchmarking, validation matrix
Wing Design	LLT & VLM 3D wing analysis
Stability Analysis	$Cm\alpha$, neutral point, static margin
Sustainability	Direct ZEROe / CATNLF alignment

[1] Abbott & von Doenhoff, Theory of Wing Sections (Dover, 1959) · [2] NACA Report 824 — 6-Series Airfoils · [3] UIUC Airfoil Coordinate Database · [4] Airbus ZEROe Sustainable Aviation Programme · [5] NASA CATNLF Laminar Flow Research · [6] AIAA Top Aerospace Technologies 2026